Audit of the architecture figure and the algorithms note
against TrustRegionRadius.jl

Abstract

We audit `Article3/fig-architecture.tex` and `Article3/old_versions/TRR-algorithmes.tex` against the package source. Thirty-five claims were checked. Twenty-six are discrepancies, of which fourteen are wrong rather than merely out of date or incomplete. Three of the fourteen would mislead a reader about what the code does: the algorithm accepts a step on η_1 where the package accepts on η, the R-grad pseudo-code has two branches where the code has four and uses the old γ numbering, and the R-RTR pseudo-code omits the acceptance branch that the code calls mandatory. Both documents have been corrected and both compile.

Contents

1 Method

Every claim was checked against a file in the package, not against a docstring where the code could be read instead. Counts were taken by loading the package and querying it rather than by reading export lists. The two documents were then corrected, each edit asserting that it matched its target exactly once so that a silent no-op was impossible.

Table 1: What the package actually contains, counted by loading it.

axis	count	members
radius rules	**11**	`RDelta`, `RStep`, `RDeltaStep`, `RDFO`, `RGrad`, `RGradCapped`, `RAdaptiveStep`, `RAdaptiveGrad`, `RAdaptiveGradCapped`, `RRTR`, `RRTRGrad`
of those, criticality-anchored	**6**	`RDFO`, `RGrad`, `RGradCapped`, `RAdaptiveGrad`, `RAdaptiveGradCapped`, `RRTRGrad`
τ variants exported	**6**	one per criticality-anchored rule
model Hessians	**8**	`ExactHessian`, `LBFGSModel`, `SR1Model`, `ScaledIdentity`, `SPDTarget`, `BHHHModel`, `BHHH2Model`, `GaussNewtonModel`
subproblem solvers	**5**	`SteihaugCG`, `ExactMS`, `KrylovCG`, `KrylovCR`, `KrylovCGLanczos`
plus one wrapper	**1**	`EigenPoint`
sampling rules	**10**	`FullBatch` to `CertifiedDecrease`
experiment files	**16**	`exp1` to `exp17`, no `exp10`

2 The three errors that matter

These are wrong rather than out of date, and each would lead a reader to a different algorithm from the one implemented.

2.1 Acceptance was attributed to the wrong threshold

The algorithm read

```
\If{$\rho \ge \eta_1$}{      % acceptance
```

and declared its inputs as $0 \le \eta_1 < \eta_2 < 1$, so η did not appear at all. The package has three thresholds and accepts on the first of them. `src/Trust-region/common.jl:562` reads

```
st.accepted = st.ρ >= p.η    # acceptance is decided by η alone
```

and `common.jl:161` enforces $0 \le \eta \le \eta_1 \le \eta_2 < 1$.

This is not a notational difference. With $\eta < \eta_1$ the band $\rho_k \in [\eta, \eta_1)$ opens, in which the step is accepted and the radius nonetheless contracts. A document that accepts on η_1 cannot express that band, and the band is the subject of an entire experiment, `exp14_acceptance_band.jl`. The default $\eta = \eta_1$ reproduces the coupled behaviour, so the old text describes the default rather than the rule.

2.2 R-grad had two branches and the old γ numbering

The pseudo-code read

```
if rho >= eta2 and |s| > Delta/2 :  mu <- gamma2 * mu
elseif rho < eta1 :                 mu <- gamma1 * mu
return mu * |g_{k+1}|
```

The code at `src/Radius_updates/rules.jl:516--529` has four branches, and expansion uses γ_3:

```
if      rho <  eta1                        : mu <- gamma1 * mu    (contract)
elseif rho <  eta2                         : mu <- gamma2 * mu    (shrink)
elseif !half_test || |s| > Delta/2         : mu <- gamma3 * mu    (expand)
else                                       : mu unchanged         (hold)
```

The missing middle branch is a mild contraction on a merely successful step, and the old text sends that case to the *expansion* branch by omission. The γ_2 for expansion is the numbering before the rule was renumbered; `benchmark/config.jl` records the change and warns that a positional call can pass silently under the wrong reading.

2.3 R-RTR omitted the branch the code calls mandatory

The pseudo-code branched on $\widetilde{\rho}$ alone. The code at `rules.jl:993--1010` branches on the acceptance flag first, and its docstring says why:

> Branching on `accepted` is mandatory, not defensive. Comparing a *classical* ρ against the *retrospective* thresholds would leave the radius untouched for $\rho \in [\widetilde{\eta}_1, \eta)$, an unsuccessful iteration with no contraction, which forfeits weak admissibility and, since a rejected step changes neither the model nor the iterate, reproduces the same subproblem for ever.

The old pseudo-code also used γ_2 where the code uses γ_3, and γ_1 where the code uses γ_2, and invented a γ_0 that does not exist. The correct rule is

$$\Delta_{k+1} = \begin{cases} \gamma_2 \|s_k\| & \text{rejected, } \rho \geq 0, \\ \min\{\gamma_2 \|s_k\|, \gamma_1 \Delta_k\} & \text{rejected, } \rho < 0, \\ \max\{\gamma_3 \|s_k\|, \Delta_k\} & \text{accepted, } \widetilde{\rho} \geq \widetilde{\eta}_2, \\ \Delta_k & \text{accepted, } \widetilde{\eta}_1 \leq \widetilde{\rho} < \widetilde{\eta}_2, \\ \gamma_2 \|s_k\| & \text{accepted, } 0 \leq \widetilde{\rho} < \widetilde{\eta}_1, \\ \min\{\gamma_2 \|s_k\|, \gamma_1 \Delta_k\} & \text{accepted, } \widetilde{\rho} < 0. \end{cases}$$

3 The full audit

3.1 `fig-architecture.tex`

claim	verdict	what the package has	source
"nine mechanisms"	**wrong**	eleven	`TrustRegionRadius.jl:50--52`
"each with a second-order variant"	**wrong**	only the six criticality-anchored ones; `SecondOrder` raises `ArgumentError` on the others	`Second_order/second_order.jl:134`
eight model Hessians	ok	eight	`TrustRegionRadius.jl:67--68`
"Steihaug–Toint CG, Moré–Sorensen, Krylov CG/CR, eigenpoint"	partial	`KrylovCGLanczos` missing; `EigenPoint` is a wrapper and not a fifth solver	`TrustRegionRadius.jl:74`
ten sampling rules	ok	ten	`TrustRegionRadius.jl:79--82`
"one driver, shared"	partial	the *step* is shared by three solvers; the loops around it differ	`Trust-region/common.jl:478`

claim	verdict	what the package has	source
Step 1 = "Build m_k"	partial	Step 1 also evaluates ω_k and, when needed, $\lambda_{\min}$	`common.jl:430--431`
Step 3 = "ρ_k, accept"	partial	does not say the threshold is η and not η_1	`common.jl:562`
per-iteration record, ten fields	partial	the trace carries eighteen; f_k, accepted, γ_k, ξ_k, τ_k were omitted	`common.jl:672--692`
diagnostics, six	partial	nine are exported; θ_k and `hypotheses_report` omitted	`TrustRegionRadius.jl:106--108`
benchmark, four	ok	all four are exported	`TrustRegionRadius.jl:111--112`

3.2 `TRR-algorithmes.tex`

claim	verdict	what the package has	source
"deux solveurs du sous-problème"	**wrong**	five, plus `EigenPoint`	`Subproblem/subproblem.jl`
"dix mécanismes"	**wrong**	eleven	`rules.jl`
"sept expériences"	**stale**	sixteen files, `exp1` to `exp17`	`benchmark/experiments/`
"trois axes"	**wrong**	four; the oracle is the fourth and the note describes it nowhere	`TrustRegionRadius.jl:77--92`
RAdaptiveFanYuan	**wrong**	no such rule; the third adaptive rule is `RAdaptiveGradCapped`	`rules.jl:847`
`RDeltaStep` absent	**stale**	exported and implemented	`rules.jl:792`
models: five listed	**stale**	eight; BHHH, BHHH-2, Gauss–Newton missing	`Model_Hessians/model_hessian.jl`
input $0 \leq \eta_1 < \eta_2 < 1$	**wrong**	$0 \leq \eta \leq \eta_1 \leq \eta_2 < 1$, three thresholds	`common.jl:161`
accept on $\rho \geq \eta_1$	**wrong**	$\rho \geq \eta$	`common.jl:562`
stopping test $\|g\| \leq \epsilon$	partial	also $\lambda_{\min} \geq$ −`tol_H` when the second-order test is on, and the status is then `second_order`	`common.jl:623`
`update_radius` takes eight arguments	**wrong**	nine; the `accepted` flag was omitted	`rules.jl:307`
last two arguments are $\|g_k\|$, $\|g_{k+1}\|$	**wrong**	they are ω_k, ω_{k+1}, which are τ under `SecondOrder`	`common.jl:608`
`initial_radius` clamped to $\Delta_{\max}$	partial	clamped to $[\Delta_{\min}, \Delta_{\max}]$	`common.jl:431`
R-step contracts on $\gamma_1\Delta$	**wrong**	$\gamma_1 \|s\|$ by default; `contract_on_step` is `true`	`rules.jl:363 and 377`
R-grad two branches, γ_2 expands	**wrong**	four branches, γ_3 expands	`rules.jl:516--529`
R-RTR no acceptance branch, γ_0	**wrong**	branches on `accepted` first; no γ_0 exists	`rules.jl:993--1010`
R-DFO compares against $\|g_k\|$	partial	against ω_k	`rules.jl:438`
only `first_order` counts as solved	**wrong**	`second_order` counts too	`benchmark/harness.jl:191`

claim	verdict	what the package has	source
Steihaug pseudo-code	ok	matches, including the three exits	`subproblem.jl:193--246`
Moré–Sorensen pseudo-code	ok	matches, hard case included	`subproblem.jl:291`
retrospective ratio, sign of the quadratic term	ok	$-g_{k+1}^\top s_k + \frac{1}{2} s_k^\top H_{k+1} s_k$, returns $-\infty$ when non-positive	`retrospective.jl:48--51`
`SPDTarget` condition $\varphi(x) < 0$	ok	matches `phi_target`	`model_hessian.jl:352`
stagnation guard	ok	matches, including both conditions	`common.jl:553`
performance and data profiles	ok	match	`Benchmark/profiles.jl`

Table 4: Thirty-five claims by severity, counted from the two tables above. Eleven are from the figure and twenty-four from the note.

severity	count	meaning
wrong	14	the document states something the code does not do
stale	3	true when written, overtaken by later work
partial	9	correct as far as it goes, incomplete
ok	9	checked and correct
total	35	of which 26 are discrepancies

4 What was changed

4.1 `fig-architecture.tex`

Six edits. The radius-rule box now reads "eleven mechanisms, the six criticality-anchored ones with a variant $\omega_k = \tau_k$". The subproblem box gains Krylov CG-Lanczos and names `EigenPoint` as a wrapper. Step 1 becomes "Build m_k, ω_k" and Step 3 "ρ_k, accept if $\rho_k \geq \eta$". The loop label becomes "one iteration, shared by the three problem classes". The record gains f_k, accepted, γ_k, ξ_k and τ_k; the diagnostics box gains θ_k. The caption gains one sentence saying that Step 1 evaluates the measure and that Step 3 accepts on η while Step 4 scales on η_1 and η_2.

This file is `\input` by `Survey-part3-v1.tex` at line 599, so the change reaches the paper.

4.2 `TRR-algorithmes.tex`

Twenty-three edits, each asserted to match once. The three of Section 2, the counts, the axes table, the model table, the experiment table extended from seven rows to sixteen, the success criterion, and a new paragraph (b′) explaining that acceptance and scaling are two decisions. The note compiles to ten pages with no errors.

Table 5: The corrected file compiles.

file	result	pages
`fig-architecture.tex`	`\input` only, no standalone build	n/a
`TRR-algorithmes.tex`	0 errors	10

5 What was not changed, and why

The location of the algorithms note. `TRR-algorithmes.tex` sits in `Article3/old_versions/`. We corrected it in place because that is what was asked, but a note describing the current package does not belong in a directory named for superseded drafts. The directory holds four other superseded drafts of the numerical study. Moving it is a decision about the repository and not about the mathematics, so we leave it.

The note is in French and the study is in English. We kept the language. Changing it is an editorial decision.

The five families. The section heading "Les cinq familles" still shows five representative rules out of eleven. That is a reasonable exposition and not an error, so it stands. The axes table now gives the full list, so the count is available.

The bibliography. `TRR-algorithmes.tex` points at `\bibliography{../References}` and compiles without it, so the citation keys were not checked against `References.bib`. That is a separate audit.

6 Discussion

The errors cluster where the package changed. Most of the wrong claims concern things the package gained after the note was written: the third threshold η, the criticality measure ω_k in place of $\|g_k\|$, the `accepted` flag, the second-order stopping test, and the rules and models added since. The Steihaug and Moré–Sorensen pseudo-code, which describes parts that have not moved, is correct line by line. That is the expected pattern and it says where to look next time.

One error is of a different kind. The γ numbering in R-grad and R-RTR is wrong because the rules were renumbered, and `benchmark/config.jl` carries a comment warning that a positional call can pass silently under the wrong reading. The note is exactly that kind of silent reader. A document that states γ_2 where the code means γ_3 produces plausible radii rather than a visible failure, which is the same hazard the comment describes.

The acceptance threshold is the one to fix first. Of the three errors of Section 2, the acceptance threshold is the one that changes what a reader would implement. The other two are wrong about branches within a rule; this one is wrong about the trust-region method itself, and it removes the degree of freedom that `exp14_acceptance_band.jl` exists to measure.